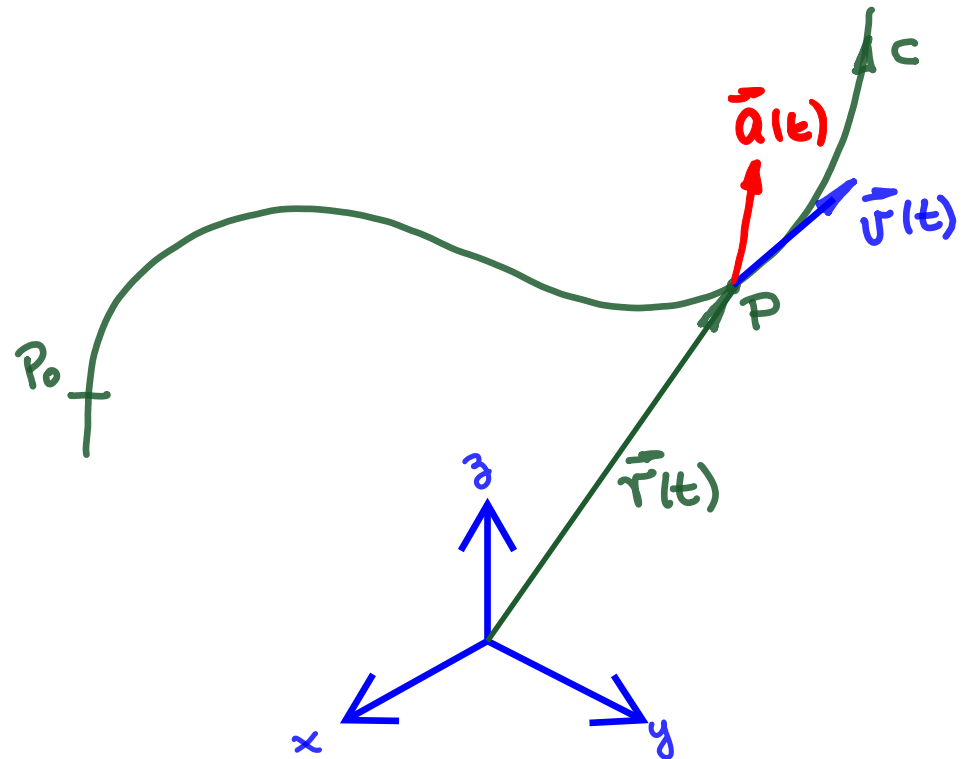


Tangential and Normal Components of Acceleration

Trim Ch 11, S. 11.13

Outline

1. Tangential and Normal Components of Acceleration
2. A different formula for Curvature



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Tangential and Normal Components of Acceleration

For a body traveling in a curve defined by the path $\vec{r}(t)$ that is accelerating, the acceleration is given by the derivative of the velocity vector.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{d\vec{r}}{dt} \right)$$

Using the chain rule, we can write the velocity vector as

$$\vec{v} = \frac{d\vec{r}}{ds} \cdot \frac{ds}{dt}$$

$$\vec{v} = \hat{T} \cdot \frac{ds}{dt}$$

DIFFERENTIATING BOTH SIDES ,

$$\frac{d\vec{v}}{dt} = \vec{a} = \frac{d}{dt} \left(\hat{T} \frac{ds}{dt} \right)$$



Tangential and Normal Components of Acceleration

$$= \underbrace{\frac{d\hat{T}}{dt} \cdot \frac{ds}{dt}}_{\textcircled{I}} + \underbrace{\frac{d}{dt} \left(\frac{ds}{dt} \right)}_{\frac{d^2s}{dt^2}} \cdot \hat{T}$$

USING CHAIN RULE IN $\textcircled{I}$

$$\frac{d\hat{T}}{dt} \cdot \frac{ds}{dt} = \left[\frac{d\hat{T}}{ds} \cdot \frac{ds}{dt} \right] \cdot \frac{ds}{dt}$$

$$= \left(\frac{ds}{dt} \right)^2 \frac{d\hat{T}}{ds} = \kappa \hat{N} \left(\frac{ds}{dt} \right)^2$$

$$\hat{N} = \frac{1}{\kappa} \frac{d\hat{T}}{ds}$$

$$\vec{a} = \frac{d^2s}{dt^2} \hat{T} + \kappa \left(\frac{ds}{dt} \right)^2 \hat{N}$$



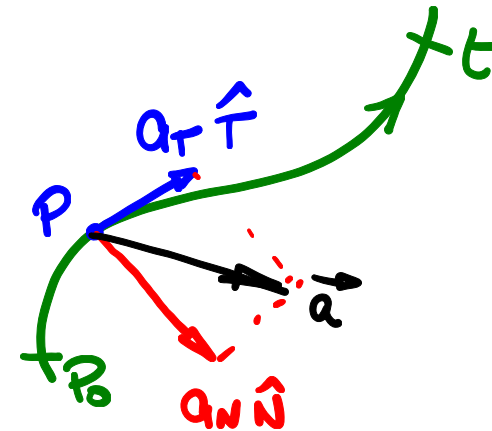
Acceleration components in terms of “t”

Tangential scalar component a_T :

$$a_T = \frac{d^2s}{dt^2} = \frac{d}{dt} \underbrace{\left(\frac{ds}{dt} \right)}_{|\vec{v}|} \rightarrow a_T = \frac{d|\vec{v}|}{dt}$$

Normal scalar component a_N :

$$a_N = \kappa \underbrace{\left(\frac{ds}{dt} \right)^2}_{|\vec{v}|^2} = \kappa |\vec{v}|^2$$



Notice that a_N makes the particle/body follow a curved path with direction parallel to the instantaneous radius of curvature “ ρ ” (centripetal force).



Another formula for the Normal Component of Acceleration

Another formula for the normal component of the acceleration, which does not require us to calculate the curvature ($k = \frac{1}{|\vec{v}|} \frac{d\hat{T}}{dt}$), can be obtained from using the fact that,

$$|\vec{a}|^2 = \vec{a} \cdot \vec{a}$$

$$|\vec{a}|^2 = (a_T \hat{T} + a_N \hat{N}) \cdot (a_T \hat{T} + a_N \hat{N})$$

$$|\vec{a}|^2 = a_T^2 + a_N^2$$

$$\Rightarrow a_N^2 = |\vec{a}|^2 - a_T^2 \quad \Rightarrow \quad a_N = \sqrt{|\vec{a}|^2 - a_T^2}$$

$$\vec{a} = a_T \hat{T} + \underbrace{\sqrt{|\vec{a}|^2 - a_T^2}}_{a_N} \hat{N}$$



Another Formula for Curvature

Find a different formula for curvature using the following formulas for velocity,

$$\vec{v} = \frac{ds}{dt} \hat{T}$$

and for acceleration,

$$\vec{a} = \frac{d^2s}{dt^2} \hat{T} + k \left(\frac{ds}{dt} \right)^2 \hat{N}$$

Solution:

LET US WRITE :

$$\vec{v} \times \vec{a} = \left(\frac{ds}{dt} \right) (\hat{T}) \times \left[\frac{d^2s}{dt^2} \hat{T} + k \left(\frac{ds}{dt} \right)^2 \hat{N} \right]$$



Another Formula for Curvature

$$= \frac{ds}{dt} \left(\frac{d^2s}{dt^2} \right) \left(\hat{T} \times \hat{T} \right) + \kappa \left(\frac{ds}{dt} \right)^3 \underbrace{\hat{T} \times \hat{N}}_{\hat{B}}$$



TAKING THE MAGNITUDE

$$|\vec{v} \times \vec{a}| = \underbrace{\left| \frac{ds}{dt} \right|^3}_{|\vec{v}|^3} \underbrace{|\hat{B}|}_{=1} \rightarrow |\vec{v} \times \vec{a}| = \kappa |\vec{v}|^3$$

$$\Rightarrow \boxed{\kappa = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3}}$$

VECTOR FORMULA
FOR κ

$$\rightarrow \rho = \frac{|\vec{v}|^3}{|\vec{v} \times \vec{a}|}$$

RADIUS
OF CURV.



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Problem

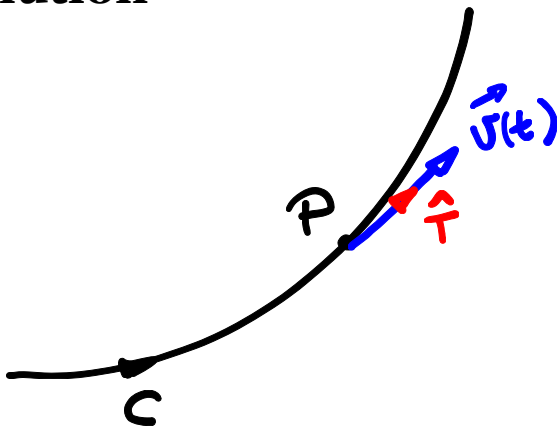
A particle has a trajectory defined by,

$$x = t, \quad y = t^2, \quad z = t^2$$

where “ t ” is time, $t \geq 0$

Find the **normal and tangential components** of its velocity and acceleration and express them in terms of $\hat{T}$ and $\hat{N}$, **without calculating neither $\hat{T}$ nor $\hat{N}$**

Solution



$$J_N = 0$$

(THE DIRECTION OF
MOTION AT P
IS TANGENT TO 'C',
ALWAYS MOVING ALONG
THE PATH)



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Problem

TANGENTIAL COMP. OF VELOCITY,

$$\vec{v} = \underbrace{v_T}_{|\vec{v}|} \hat{T}$$

$$\vec{r}(t) = (t, t^2, t^2) \rightarrow \vec{v}(t) = (1, 2t, 2t)$$

$$|\vec{v}(t)| = \sqrt{1 + 4t^2 + 4t^2}$$

$$|\vec{v}(t)| = \sqrt{1 + 8t^2}$$

$$v_T = \sqrt{1 + 8t^2}$$

$$\vec{v} = \sqrt{1 + 8t^2} \hat{T} + 0 \hat{N}$$

TANGENTIAL COMP. OF ACC.

$$a_T = \frac{d}{dt} |\vec{v}(t)| = \frac{d}{dt} \sqrt{1 + 8t^2}$$



Problem

$$Q_T = \frac{16t}{2\sqrt{1+8t^2}} = \frac{8t}{\sqrt{1+8t^2}}$$

FOR a_N ,

$$a_N = \sqrt{|\vec{a}|^2 - a_T^2} \quad ; \quad \vec{a}(t) = \frac{d\vec{v}}{dt} = (0, 2, 2)$$

SINCE:

$$\vec{a} = (0, 2, 2) \rightarrow |\vec{a}|^2 = (\sqrt{4+4})^2 = 8$$

$$a_N = \sqrt{8 - \left(\frac{8t}{\sqrt{1+8t^2}} \right)^2}$$

$$\dots \quad a_N = \frac{2\sqrt{2}}{\sqrt{1+8t^2}}$$

$$\vec{a} = \frac{8t}{\sqrt{1+8t^2}} \hat{T} + \frac{2\sqrt{2}}{\sqrt{1+8t^2}} \hat{N}$$



Summary

- The tangential component of acceleration gives us information on the rate of change of speed
- The normal component of acceleration is related to the centripetal force
- The components of the acceleration lie in the plane determined by $\hat{T}$ and $\hat{N}$, or the osculating plane